

SQL Learning Journal #1 – Introduction to SQL

1. SQL Commands Learned Today

a. Understanding SQL Tables

Data is stored in **databases**, which contain one or more **tables**.

Each table consists of:

- Rows (records or observations)
- Columns (fields)

When naming columns, it is generally recommended to use **singular nouns**, since each column represents a single attribute of a record.

Table data can have different data types, including:

- String (text)
- Integer
- Float (decimal numbers)

Most databases contain multiple related tables that share common information through relationships. Such databases are called **relational databases**.

b. Selecting Data

The `SELECT . . . FROM` statement is used to retrieve one or more columns from a table.

Select a single column:

SELECT field_name

FROM table_name;

Example:

SELECT name

FROM table_name;

Select multiple columns:

SELECT name, num

FROM table_name;

SELECT name, num, surname

FROM table_name;

Select all columns:

SELECT *

FROM table_name;

The asterisk (*) is called the **wildcard character** because it selects every column in the table.

Note: The output returned by a SQL query is called a **result set**.

2. Learning Notes:

Today I learned the fundamental concepts of SQL and relational databases. I became familiar with the basic terminology used in SQL, including databases, tables, rows, columns, data types, and relational databases. I also practiced the first SQL commands for selecting, renaming, and retrieving unique data from database tables.